



Gennoor Tech
TRAIN. INNOVATE. BUILD.

5-DAY HANDS-ON BOOTCAMP · 40 HOURS

Data Science & Machine Learning

From data exploration to production ML pipelines

5 40 15+ 5+

DAYS HOURS HANDS-ON LABS PROJECTS

www.gennoor.com · enquiry@gennoor.com

Course Overview

This 5-day hands-on bootcamp covers everything from foundational concepts to advanced techniques, combining lectures, live demos, hands-on labs, and collaborative workshops for maximum learning impact.

Prerequisites: Basic programming experience (any language). Python covered on Day 1.

Who Is This For?


Developers


Data Analysts


Fresh Graduates


L&D Teams
Internal trainers

Learning Outcomes

- 1 **Master Python for data science** — NumPy, Pandas, Matplotlib, Seaborn
- 2 **Build supervised ML models** — regression, classification, ensemble methods
- 3 **Apply unsupervised learning** — clustering, PCA, anomaly detection
- 4 **Engineer features and** — build end-to-end Scikit-learn pipelines
- 5 **Deploy ML models** — via FastAPI and Docker

5-Day Curriculum

Interactive format with live demos, hands-on labs, and collaborative workshops. Schedule and session order are fully customizable based on your organization's needs and time zones.

01

Python for Data Science & EDA

HANDS-ON LABS + WORKSHOP

- SESSION A
- Python for Data Science
- Python essentials: data types, control flow, functions, OOP
 - NumPy arrays, Pandas DataFrames, groupby, merge, pivot tables
 - Matplotlib, Seaborn, Plotly for statistical visualization

- SESSION B
- EDA
- Exploratory Data Analysis methodology
 - Lab: Complete EDA on a customer churn dataset

02

Data Preprocessing & Feature Engineering

DEMOS + LABS + DISCUSSION

- SESSION A
- Data Preprocessing
- Missing values, outlier detection (IQR, Z-score)
 - Encoding: one-hot, label, target encoding
 - Feature scaling: standardization, normalization, robust scaling

- SESSION B
- Feature Engineering
- Scikit-learn Pipelines: chaining preprocessing + training
 - Lab: Feature engineering for a credit scoring dataset

03

Supervised Learning — Regression & Classification

LECTURE + HANDS-ON LABS

SESSION A

Supervised Learning

- Linear, Polynomial, Ridge, Lasso, ElasticNet regression
- Logistic Regression, Decision Trees, Random Forests
- Gradient Boosting: XGBoost, LightGBM

SESSION B

Regression & Classification

- Evaluation: MSE, RMSE, R², confusion matrix, ROC-AUC
- Lab: Customer churn prediction model

04

Unsupervised Learning & Model Optimization

HANDS-ON LABS + WORKSHOP

SESSION A

Unsupervised Learning

- K-Means, DBSCAN, hierarchical clustering
- PCA for dimensionality reduction
- Cross-validation, GridSearchCV, Optuna

SESSION B

Model Optimization

- Ensemble methods: bagging, boosting, stacking
- Lab: Fraud detection model optimization

05

NLP Fundamentals, Deployment & Capstone

DEMOS + LABS + DISCUSSION

SESSION A

NLP Fundamentals, Deployment

- Text preprocessing, TF-IDF, word embeddings
- Sentiment analysis with Scikit-learn
- Introduction to Hugging Face Transformers

SESSION B

Capstone

- Model serving with FastAPI + Docker
- Capstone: End-to-end ML project

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Platforms Covered

All tools demonstrated and practiced hands-on during the bootcamp

- Docker
- FastAPI
- Hugging Face
- NumPy
- Pandas
- Python
- Scikit-learn

What You Take Home



5 ML Projects (Jupyter notebooks with code & data)
Included with course completion



ML Pipeline Templates (Scikit-learn)
Included with course completion



EDA Toolkit (notebook templates)
Included with course completion



Cheat Sheets (Pandas, Scikit-learn, metrics)
Included with course completion

Certification of Completion



Gennoor Tech — Certificate of Completion

Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in "**Data Science & Machine Learning**". Digitally signed by the Founder & Director. Validates proficiency in all course topics and hands-on competency.

Recommended Next Certifications

DP-100
Azure Data Scientist

AI-901
Azure AI Fundamentals (New 2026)

AI-103
Azure AI Solutions

Designed & Delivered By

Jalal Ahmed Khan
Founder & Director, Gennoor Tech Private Limited

Senior AI Consultant and **Microsoft Certified Trainer (MCT)** with 14+ years of **16+ active Microsoft certifications** experience. Delivered AI and technology programs for Microsoft, EY, IBM, Boeing, including AB-100 (Agentic AI), AI-103, AZ-400, Saudi Aramco, HDFC Bank, and 80+ organizations across 6+ countries including the AI-300, PL-400, and GitHub certifications. 10-day AI Leadership program for MCIT Saudi Arabia.

Builder of open-source MCP servers and Copilot Studio integrations. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options  **On-Site** — customized at your office |  **Virtual Live** — interactive via Teams/Zoom |  **International** — GCC, Africa, APAC, UK



Ready to Start Your Data Science & Machine Journey?

Contact us to discuss a customized program
for your organization or register for the next open batch.

🌐 www.gennoor.com

✉ enquiry@gennoor.com

☎ +91 9326 352 241

🔗 linkedin.com/company/gennoor-tech-private-limited

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